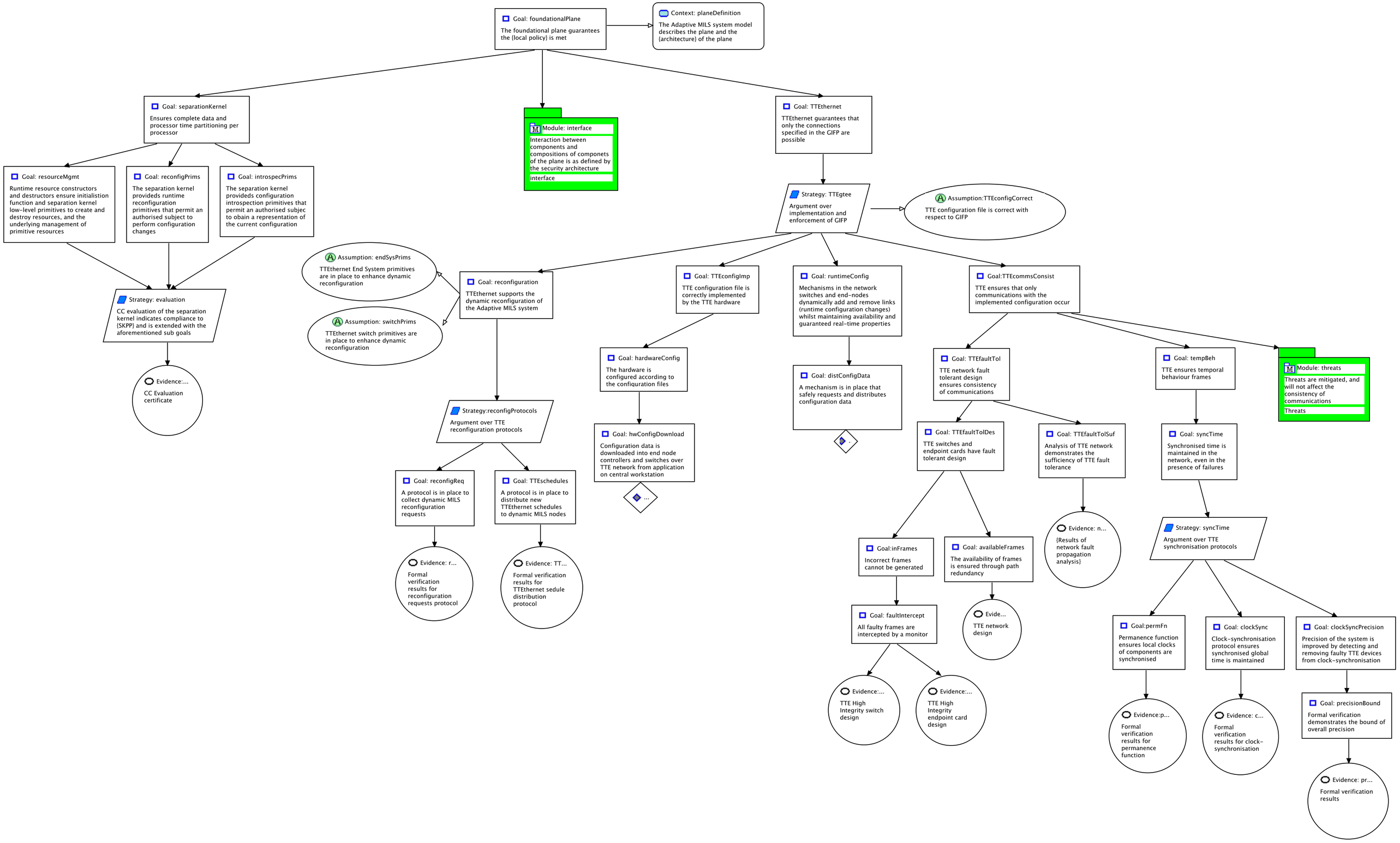
Goal: foundationalPlane
The foundational plane guarantees the {local policy} is met
Context: planeDefinition
The Adaptive MILS system model describes the plane and the {architecture} of the plane
Goal: separationKernel
Ensures complete data and processor time partitioning per processor
Module: interface
Interaction between components and compositions of componets of the plane is as defined by the security architecture
interface
Goal: TTEthernet
TTEthernet guarantees that only the connections specified in the GIFP are possible
Goal: resourceMgmt
Runtime resource constructors and destructors ensure initialistion function and separation kernel low-level primitives to create and destroy resources, and the underlying management of primitive resources
Goal: reconfigPrims
The separation kernel provideds runtime reconfiguration primitives that permit an authorised subject to perform configuration changes
Goal: introspecPrims
The separation kernel provideds configuration introspection primitives that permit an authorised subjec to obain a representation of the current configuration
Strategy: TTEgtee
Argument over implementation and enforcement of GIFP
Assumption:TTEconfigCorrect
TTE configuration file is correct with respect to GIFP
Assumption: endSysPrims
TTEthernet End System primitives are in place to enhance dynamic reconfiguration
Goal: reconfiguration
TTEthernet supports the dynamic reconfiguration of the Adaptive MILS system
Goal: TTEconfigImp
TTE configuration file is correctly implemented by the TTE hardware
Goal: runtimeConfig
Mechanisms in the network switches and end-nodes dynamically add and remove links (runtime configuration changes) whilst maintaining availability and guaranteed real-time properties
Goal:TTEcommsConsist
TTE ensures that only communications with the implemented configuration occur
Strategy: evaluation
CC evaluation of the separation kernel indicates compliance to {SKPP} and is extended with the aforementioned sub goals
Evidence:...
CC Evaluation certificate
Assumption: switchPrims
TTEthernet switch primitives are in place to enhance dynamic reconfiguration
Goal: hardwareConfig
The hardware is configured according to the configuration files
Goal: distConfigData
A mechanism is in place that safely requests and distributes configuration data
Goal: TTEfaultTol
TTE network fault tolerant design ensures consistency of communications
Goal: tempBeh
TTE ensures temporal behaviour frames
Module: threats
Threats are mitigated, and will not affect the consistency of communications
Threats
Strategy:reconfigProtocols
Argument over TTE reconfiguration protocols
Goal: hwConfigDownload
Configuration data is downloaded into end node controllers and switches over TTE network from application on central workstation
...
Goal: TTEfaultTolDes
TTE switches and endpoint cards have fault tolerant design
Goal: TTEfaultTolSuf
Analysis of TTE network demonstrates the sufficiency of TTE fault tolerance
Goal: syncTime
Synchronised time is maintained in the network, even in the presence of failures
Goal: reconfigReq
A protocol is in place to collect dynamic MILS reconfiguration requests
Goal: TTEschedules
A protocol is in place to distribute new TTEthernet schedules to dynamic MILS nodes
Evidence: r...
Formal verification results for reconfiguration requests protocol
Evidence: TT...
Formal verification results for TTEthernet sedule distribution protocol
Evidence: n...
{Results of network fault propagation analysis}
Strategy: syncTime
Argument over TTE synchronisation protocols
Goal:inFrames
Incorrect frames cannot be generated
Goal: availableFrames
The availability of frames is ensured through path redundancy
Evide...
TTE network design
Goal: faultIntercept
All faulty frames are intercepted by a monitor
Evidence:...
TTE High Integrity switch design
Evidence:...
TTE High Integrity endpoint card design
Goal:permFn
Permanence function ensures local clocks of components are synchronised
Goal: clockSync
Clock-synchronisation protocol ensures synchronised global time is maintained
Goal: clockSyncPrecision
Precision of the system is improved by detecting and removing faulty TTE devices from clock-synchronisation
Evidence:p...
Formal verification results for permanence function
Evidence: c...
Formal verification results for clock-synchronisation
Goal: precisionBound
Formal verification demonstrates the bound of overall precision
Evidence: pr...
Formal verification results